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Raj Nandan

Software Engineer - Backend | DevOps

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Backend & **DevOps engineer** with **internship experience** in **observability**, **alerting**, and production database migrations, plus a portfolio of independently **built microservice** systems covering **event-driven architecture**, multi-tenant auth, and real-time monitoring. Proficient in Java/Spring Boot, Node.js/NextJS, and **cloud-native** tooling (Docker, Kubernetes, CI/CD). Currently building **agentic AI** projects to extend into applied **Generative AI.**

Skills

Programming	Cpp (DSA), Python, Java, JavaScript, SQL
Backend & APIs / Frontend	NextJS, Node.js, Express.js, TypeScript, REST API Design, Spring Boot, Microservices, React
Databases	MySQL, PostgreSQL, MongoDB, Redis
Cloud & DevOps	Docker, Kubernetes (basic scaling/rollouts), Prometheus, Grafana, ELK Stack
CS Fundamentals	Object-Oriented Programming (OOP), Data Structures & Algorithms, DBMS, System Design

Technical Experience

Site Reliability Engineering Intern / BARQ, Bengaluru **March 2025-Aug 2025**

- **Set up and maintained** Grafana dashboards and Prometheus/ELK-based monitoring for **critical services**, and **configured database alerting** rules to proactively surface anomalies ahead of production impact
- Participated in **production deployments** and roll-out verification, and helped author incident runbooks
- Contributed to **testing and validation** of a production-to-DR (disaster recovery) **database migration** and reverse failback, verifying data integrity through the cutover
- **Assisted in deploying and scaling** containerized services on Docker and Kubernetes

Full Stack Developer / Freelance, Remote **Jan 2024 – Dec 2024, alongside final year of B.Tech**

- **Built and shipped** multiple **MERN-stack applications** — dashboards, client tools, and workflow automation — for freelance clients
- **Designed** REST APIs with Node.js, Express, and MongoDB, and **applied Redis caching** to reduce repeated query load
- **Improved application** performance using caching strategies and efficient state/data handling on both **front-end and backend**
- **Integrated AI-assisted dev tools** (GitHub Copilot, Cursor) into daily workflow to speed up iteration

Technical Projects

TaskFlow — Distributed Background-Job System / Java 21, Spring Boot 3, PostgreSQL, RabbitMQ, MinIO, Docker, React, Prometheus, Grafana, OpenTelemetry

- **Designed** an async accept-then-process platform (**React frontend, stateless API, worker pool**) where the API validates, persists, and queues report jobs in milliseconds (**202 + jobId**) while workers drain a **RabbitMQ** queue and store CSV/PDF results in **MinIO** behind short-lived signed URLs
- **Guaranteed correctness under concurrency** with a **PostgreSQL atomic claim** (`UPDATE...WHERE status='QUEUED'`, exactly one worker wins), **idempotent** job submission via `(tenant_id, idempotency_key)` unique constraint, **tenant isolation** derived server-side from API keys, and worker **lease/heartbeat** recovery for crashed claims
- **Proved scale and observability**: **k6** load tests (100 VUs, 0% errors; 500 VUs, diagnosed Tomcat thread exhaustion at ~200 threads), plus full-stack telemetry — **Micrometer/Prometheus** metrics with custom job timers, **OTLP traces** to Tempo, **Promtail** logs to Loki, and provisioned **Grafana** dashboards covering latency, saturation, and pipeline health

Incident Management System / NestJS, TypeScript, React, PostgreSQL, MongoDB, Redis, Docker
incident-management-system

- Built an **event-driven incident platform** with a high-throughput /signals ingestion endpoint using **Redis Streams**, with debouncing and deduplication to prevent incident flooding under bursty failure signals
- **Modeled incident** lifecycle as a state machine (**OPEN → INVESTIGATING → RESOLVED → CLOSED**) requiring a structured Root Cause Analysis before closure
- Used polyglot persistence (**PostgreSQL, MongoDB, Redis**) and a React dashboard with Server-Sent Events for **live incident updates**

Other projects: AI-Powered Email Writer (Spring Boot + Gemini API), Kafka event pipeline, Go/Ollama CRUD API with local LLM summaries, uptime checker, DevOps work — GitHub/Raj-nandan

Education

Bachelor of Technology in Information Technology , 7.5 CGPA
College of Engineering Roorkee 2021-2025

CERTIFICATIONS & ACHIEVEMENTS

• **LeetCode**: 500+ solved, **1469 rating** | **GeeksforGeeks**: 500+ solved, **1789 rating**
• **2nd Place** – NEO Idea Competition 2022 (Tech Innovation Project) | **3rd Place** – Intra-Department Hackathon 2023 | **3rd Rank** – Uttarakhand Tech Fest 2023
• **Coordinator – Robotics & Innovation Club (TRAIC)**: Led Technical workshops, **mentored 50+ students**, organized inter-college tech events